

The brain

WHEN HUMAN WARRIORS ATE THE BRAINS of their admired enemies to gain their cunning and wisdom, they identified the right organ. Since Plato and Hippocrates in ancient Greece, the brain has been seen as the seat of intelligence and the soul. One person who disagreed was Aristotle, who believed the soul's home is the heart. His studies on animals correctly showed that the living brain, unlike the skin and other organs, can feel nothing. Aristotle concluded it could not be the site of consciousness. Herophilus of Alexandria, about 300 BC, carried out some of the first recorded studies of the human brain. He identified its main parts, like the cerebrum and cerebellum. He also brought the organ back to prominence as the seat of thought and intelligence. Its major anatomy was established by Renaissance times, and its basic microscopic structure by the 1900s. The marvel of the brain is the way its 100 billion nerve cells interlink via untold trillions of connections.



HOLE IN THE HEAD

Phineas Gage was foreman of a quarrying gang in the USA. In 1848 a gunpowder accident blew a metal rod through his cheek, up through the left frontal lobe of his brain, and out of his skull (above). He lived, the wound healed, but his personality changed from contented and considerate to obstinate, moody, and foul-mouthed. He was living proof that the front of the brain is involved in aspects of personality.

THE BRAIN SEEN FROM BELOW

The brain's three major parts are the brain stem, the cerebellum, and the cerebrum. In humans, the cerebrum is divided into two hemispheres that form nine-tenths of the brain's volume.

The brain is only one-fiftieth of the body's weight, yet it receives about one-fifth of the blood supply (pp. 30-33). Arteries to the brain form a ring called the circle of Willis, which may be a bypass route should one vessel become blocked.



SITE OF SPEECH

Pierre Paul Broca (1824-1880) was a surgeon, anatomist, and anthropologist. He graduated in medicine, became a professor of surgery in Paris, and was also interested in higher mathematics. He found that a small area on the outer layer of the cerebrum is responsible for coordinating the muscles in the voicebox and neck that produce speech. This is now called Broca's area, or the speech motor center. Broca also studied cancers and developed instruments for measuring brains and skulls.

